

生态位模型的原理与应用（三）

THE PRINCIPLES AND APPLICATIONS OF ECOLOGICAL NICHE MODEL

中科院动物所 乔慧捷

2026-9

如何在研究工作中使用生态位模型

Novel methods improve prediction of species' distributions from occurrence data

Jane Elith*, Catherine H. Graham*, Robert P. Anderson, Miroslav Dudík, Simon Ferrier, Antoine Guisan, Robert J. Hijmans, Falk Huettmann, John R. Leathwick, Anthony Lehmann, Jin Li, [Lucia G. Lohmann](#), Bette A. Loiselle, Glenn Manion, Craig Moritz, Miguel Nakamura, Yoshinori Nakazawa, Jacob McC. M. Overton, A. Townsend Peterson, Steven J. Phillips, Karen Richardson, Ricardo Scachetti-Pereira, Robert E. Schapire, Jorge Soberón, Stephen Williams, Mary S. Wisz, Niklaus E. Zimmermann ... [See fewer authors](#) ^

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中国科学院研究生院 博士学位论文

物种潜在分布区模型构建与评价方法

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Huijie Qiao, Jorge Soberón, Andrew Townsend Peterson 

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The n -dimensional hypervolume

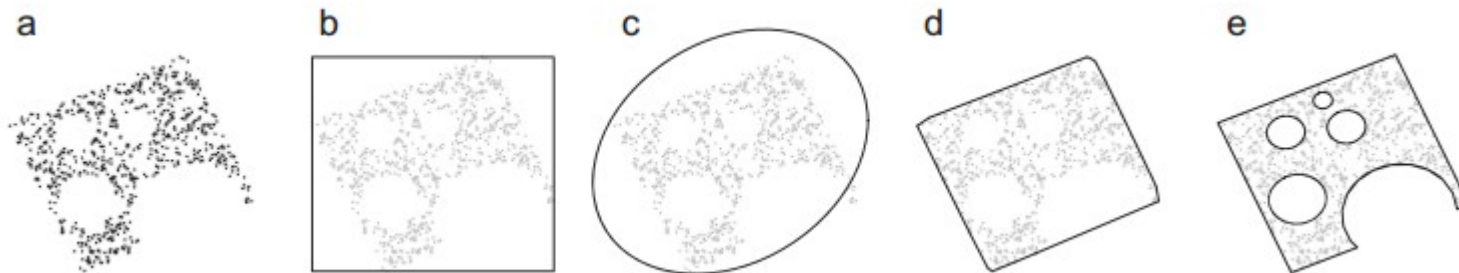
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Using the KDE method to model ecological niches: A response to Blonder et al. (2017)

Huijie Qiao, Luis E. Escobar , Erin E. Saupe, Liqiang Ji, Jorge S. S. 

First published: 11 July 2017 | <https://doi.org/10.1111/geb.12>



生态位模型前沿应用

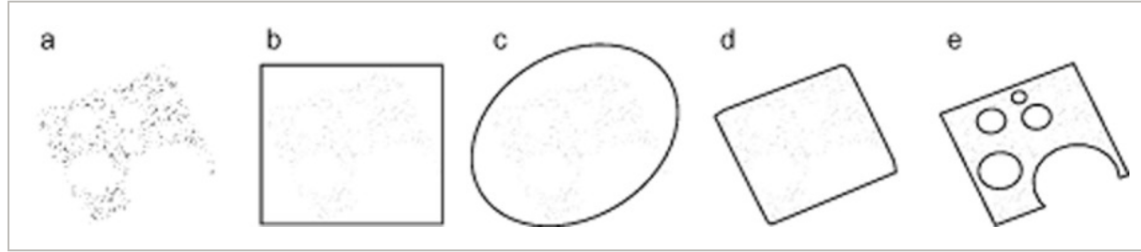


Figure 1

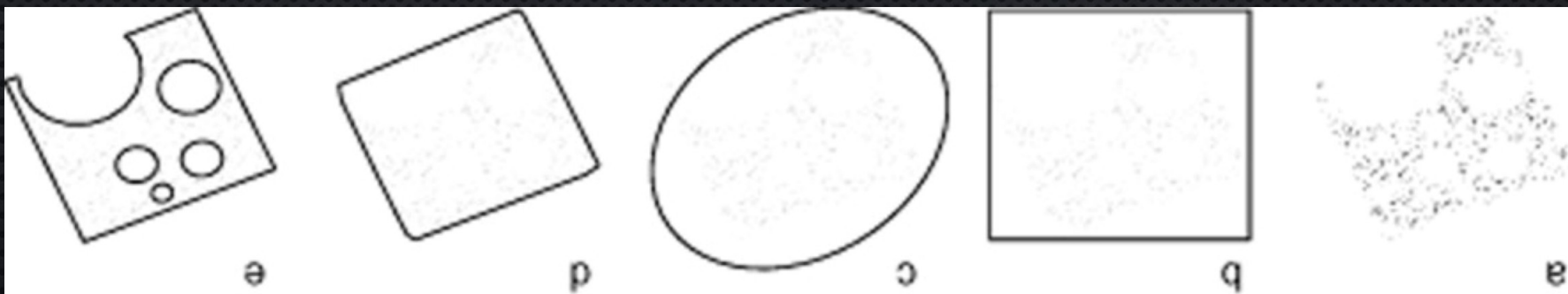
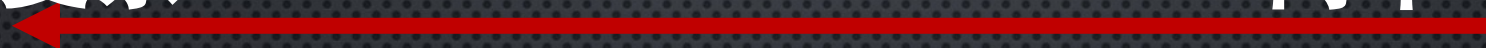
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A robust operational definition of the hypervolume is important for making correct inferences. Here we show an example of three poor definitions (b–d) and one accurate definition (e). (a) Consider a two-dimensional dataset describing a hypothetical ‘Swiss cheese’ hypervolume. (b) A two-dimensional range box fails to capture the rotation and holes in the hypervolume. (c) A principal components analysis has difficulties with non-normal data and does not account for the holes. (d) A convex hull also does not account for holes and is very sensitive to outlying points. (e) The best solution is to take the area enclosed by a contour of a kernel density estimate, which can account for non-normal, rotated and holey data including outliers. Some species distribution modelling approaches (e.g. generalized boosted regression models) can also approximate this shape.

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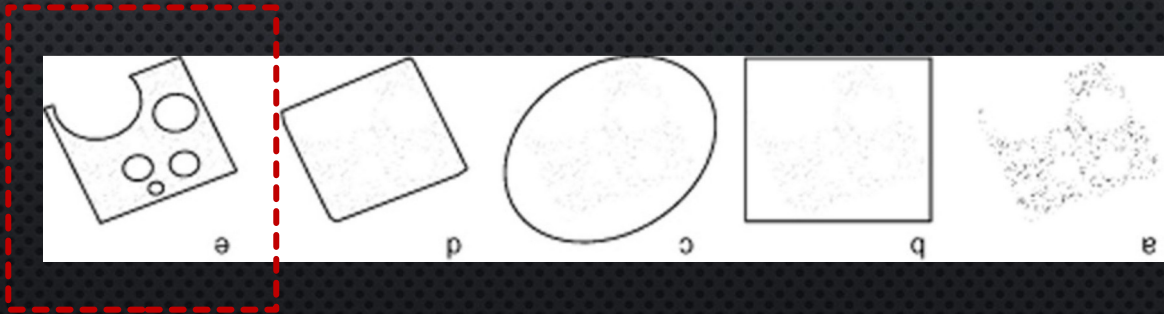
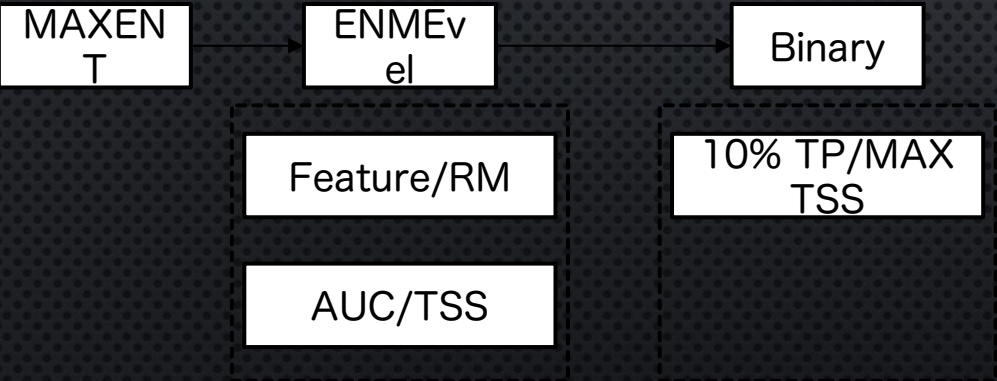
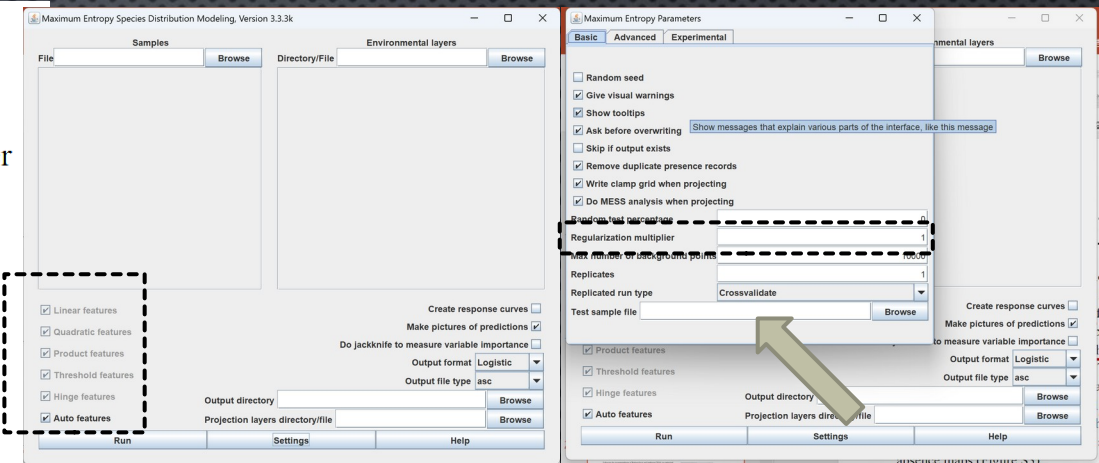
复杂

简单



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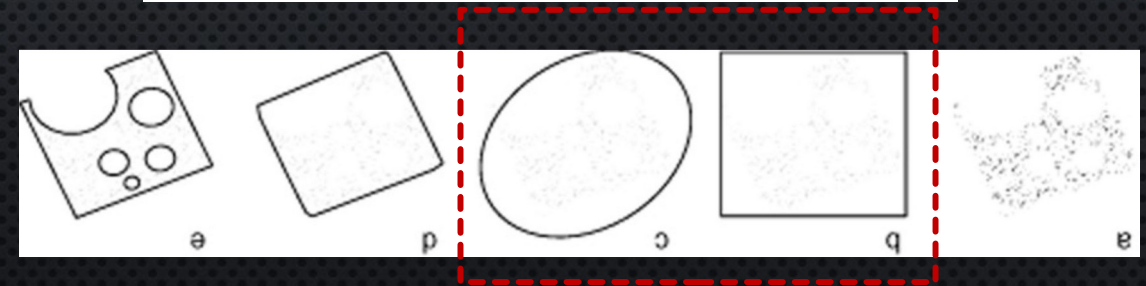
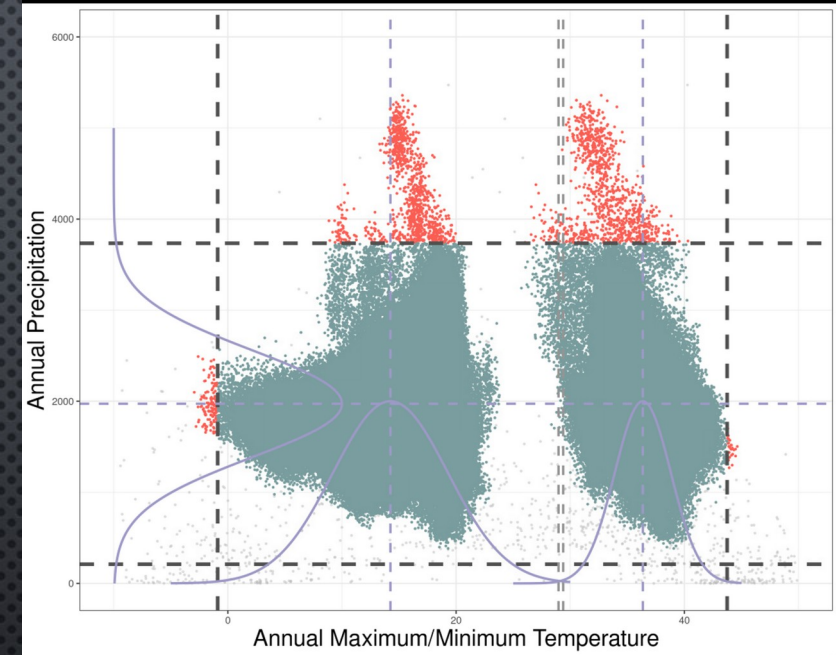
Following the recommendations of Radosavljevic and Anderson (2014), we optimized model complexity when developing MaxEnt models. In short, for each species, we first built a number of candidate models with different feature classes and regularization multipliers; from these models we then selected the optimal model via area under the curve (AUC) of the receiver operating characteristic along with the true skill statistic (TSS) (Allouche *et al.* 2006; Radosavljevic and Anderson 2014). The optimal model was then used to project species habitat suitability. Finally, continuous habitat suitability projections were converted into species presence (1) or absence (0) using a 10% presence probability threshold (Zhang *et al.* 2021) and maximizing TSS (Liu *et al.* 2013; Thuiller *et al.* 2014). Habitat suitability was quantified using the predicted richness of non-native species in each grid by summing the resultant presence-absence maps (Figure S3).



生态位模型前沿应用

Defining species-realized niche limits

Species experience variability in climatic conditions across both space and time, but this temporal variability is ignored when using time-averaged climate conditions (for example, Worldclim data⁵¹) to estimate species-realized niches. To address this, we estimated species-realized niche limits using the climate projections from the historical run of each climate model (1850–2005), which includes the influence on climate of observed changes in radiative forcing due to natural factors such as volcanic eruptions, as well as anthropogenic emissions and land-use changes⁵². Thus, in the case of MAT, we calculated the maximum MAT experienced across the species geographic range over both space and time ($T_{\max_{\text{MAT}}}$, see [Supplementary Information](#)). To prevent estimates of $T_{\max_{\text{MAT}}}$ being inflated by either extreme outliers in the temperature time series or from the overestimation of species ranges⁴⁴, we excluded outlier temperature values within each grid cell, defined as those more than three standard deviations from the mean. After we had selected the maximum temperature for each cell, we excluded outlier temperature values across each species range, defined as those more than three standard deviations above the mean range value. The $T_{\max_{\text{MAT}}}$ value for each species was then set as the maximum of the remaining values (Supplementary Fig. 1). We used an identical procedure to calculate $T_{\max}$ using MMT ($T_{\max_{\text{MMT}}}$). For precipitation, species may be exposed to either drying or wetting conditions and so we calculated both the minimum ($P_{\min}$) and maximum ($P_{\max}$) precipitation values experienced by each species across its geographic range (see [Supplementary Information](#)).



生态位模型前沿应用

Defining climatic tolerances

Grid cell occupation was controlled by species' abiotic climatic tolerances and dispersal ability in relation to the environmental conditions manifested in that cell. The temperature and precipitation values of the initial starting grid cell defined the centre of that particular simulated species' fundamental ecological niche. Because simulations began under conditions representative of 120 ka, the simulated species could be considered as 'warmer adapted' in a Pleistocene-to-recent context. We applied symmetrical deviations to the values of the initial cell based on two niche breadths (narrow and broad), corresponding to temperature and precipitation tolerance ranges of 40 °C and 60 °C, and 5 mm d⁻¹ and 10 mm d⁻¹, respectively. Niche breadths were based on plant tolerances and represent the 20% and 80% quantiles of tolerances in the Food and Agriculture Organization of the United Nations database⁵⁷. These basic tolerances (that is, fundamental ecological niches) were reduced by the set of environmental combinations existing and accessible to the species at a given time step, referred to as the existing ecological niche⁶. Occupation of the existing ecological niche at any given time step was reduced still further by the species' dispersal ability.





Climatic niche and geographic distribution

We modeled the evolution of the fundamental climatic niche (114) of populations, which defines the extremes of temperature and precipitation that a population can tolerate at any given time step (Movie 2 and fig. S7). Thus, cells occupied by a population must have climatic conditions within the limits of its fundamental niche. The realized climatic niche is an emergent property of the population, defined by the climatic conditions that it actually experiences across the whole set of occupied cells (e.g., population's range). However, not every cell with suitable climatic conditions is necessarily occupied by the population (Movie 1). Indeed, the bounds of the population's evolving fundamental climatic niche may at times extend beyond the limits of its realized niche (115), owing either to dispersal limitation (additional, climatically suitable regions currently exist but cannot be reached), niche conservatism (the fundamental niche has not yet responded to the disappearance of previously suitable climates) (26), or exclusion by competing species (see below).

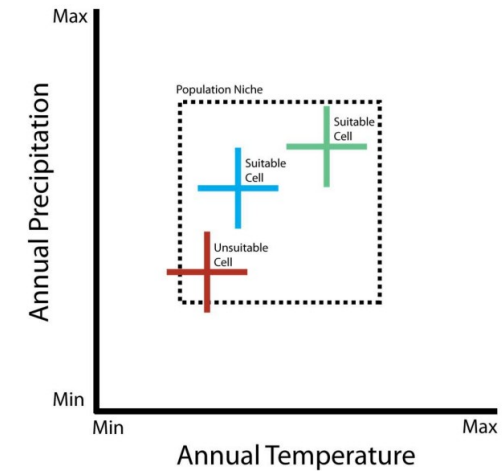


Figure S7. Graphical representation of the climatic niche of a hypothetical population (dashed square in the center), indicating minimum and maximum tolerance of temperature and precipitation, in relation to the climate in three grid cells (crosses) at a given time step of the simulation. Each cross represents a grid cell in the map, in which the horizontal length indicates the seasonality in temperature and the vertical length indicates the seasonality in precipitation. The green and blue grid cells are climatically suitable for the population, as they are enclosed within the population niche. The red grid cell is climatically unsuitable, because it is colder and drier than the limits tolerated by the population.



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- 3D/4D 厅
- 巨幕厅



SELECTED QUOTES FROM SURVEY RESPONDENTS

"The research question and the data should be king, with an approach being selected on the basis that it is appropriate to both the research question and the data rather than the research question and the data being selected to fit the approach which a person knows how to use."

"I regularly see peer-reviewed articles that apply SDM incorrectly from either a statistical or inferential perspective. This is largely a user problem rather than a software problem as some people treat [SDM Software] as black boxes rather than inferential tools, and thus do not put in the intellectual effort required to do good work."

"We don't need fancier software, we need people who understand ecology and the importance of multiple types of data ... The key is the ability to think in ecological terms."

